



Archaeological report: Discerning Archaeological Cereals Using Stable Isotopes

Author: David Larreina-García

Date: 15/06/2026

Repository: github.com/dplauto-cpu/Carpology

1. Introduction

This project was carried out as part of an assessed practical component of a Machine Learning course. While the focus is not on carpological research, datasets from academic research have been used, and basic guidance has been provided by a member of the Research Group **isoTOPIK·Lab** at the *University of Burgos* (UBU). This is a preliminary study that is expected to be revisited at some point from a purely archaeological perspective.

1.1 Archaeological Context

The domestication of cereals (wheat and barley) during the Neolithic revolution transformed human societies. However, identifying charred archaeological seeds to species level is seriously challenging due to degradation and carbonisation. Stable isotope analysis ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) offers an alternative: $\delta^{13}\text{C}$ indicates water stress, while $\delta^{15}\text{N}$ reflects aridity or manuring intensity.

1.2 Research Question

Is it possible to train a ML model to distinguish barley vs wheat isotopes, and integrate the data into a coherent geographical and chronological archaeological framework?

1.3 Objectives

- **Primary:** Build a supervised model to classify barley vs wheat
- **Secondary:** Identify isotopic clusters using unsupervised learning
- **Tertiary:** Create interactive maps for archaeological interpretation

2. Dataset

2.1 Source and Composition

The dataset `Med_plants.csv` (1341 records, 20 fields) merges:

- **MAIA database** (Mediterranean Archive of Isotopic dAta)
- **IsoTOPIKLab dataset** (University of Burgos)
- 1147 samples of barley and wheat, the rest of plants unused in the ML

2.2 Target Variable Distribution

Class	Count	Percentage
Barley	586	51.1%
Wheat	561	48.9%

2.3 Main Features

Feature	Type	Description
IRMS_d13C_Collagen	Continuous	Carbon isotope (‰)
d15N_Collagen	Continuous	Nitrogen isotope (‰)
Latitude_N	Continuous	Latitude (decimal degrees)
Longitude_E	Continuous	Longitude (decimal degrees)
Chronological_Period_clean	Categorical	Neolithic, Bronze Age, Iron Age
Mediterranean_Basin	Categorical	Western, Central, Eastern, Undefined

Cultural Period	Mean date (BP)	BCE/CE range (approx.)
Neolithic	≥ 8000	≥ 6000 BCE
Neolithic_Late	6000 – 7999	4000 – 5999 BCE
Chalcolithic	5200 – 5999	3200 – 3999 BCE
Early_Bronze	4200 – 5199	2200 – 3199 BCE
Middle_Late_Bronze	3500 – 4199	1500 – 2199 BCE
Iron_Age_Early	3000 – 3499	1000 – 1499 BCE
Iron_Age_Late	2400 – 2999	500 – 999 BCE
Iron_Roman	1700 – 2399	500 BCE – 250 CE

Region	Longitude (decimal degrees)	Latitude (decimal degrees)
General limits	-10 ≤ lon ≤ 42	25 ≤ lat ≤ 46
Western_Med	-10 ≤ lon ≤ 11	25 ≤ lat ≤ 46
Central_Med	11 < lon ≤ 23	25 ≤ lat ≤ 46
Eastern_Med	23 < lon ≤ 42	25 ≤ lat ≤ 46

3. Archaeological Interpretation and Results

3.1 Key Archaeological Findings

Finding 1: Isotopes alone cannot perfectly classify barley vs wheat

The ceiling of ~75% accuracy (AUC 0.824) indicates that isotopic signatures of barley and wheat overlap substantially. This is not a model limitation but reflects archaeological reality: the same species grown in different environmental conditions produces different isotopic values.

Archaeological implication: Environmental context (water availability, soil type, agricultural practices) must be considered alongside species identification.

Finding 2: Water stress ($\delta^{13}\text{C}$) is the strongest predictor

With 41% feature importance, $\delta^{13}\text{C}$ is the primary driver of classification. More negative $\delta^{13}\text{C}$ values (better water availability) are associated with:

- Western Mediterranean samples
- Neolithic and early Bronze Age periods
- Wheat more than barley (but with overlap)

Archaeological implication: Water management (irrigation, field location) may be as important as species choice in ancient agricultural systems.

Finding 3: Four distinct isotopic clusters reveal agricultural intensity gradients

The K-means clustering (k=4) reveals a continuum of agricultural conditions:

Cluster	Ecological interpretation	Dominant region	Chronological peak
0	Optimal conditions (low stress)	Western Mediterranean	Neolithic
1	Moderate stress	Central Mediterranean	Bronze Age
2	High stress	Eastern Mediterranean	Iron Age
3	Extreme aridity / manuring	Eastern Mediterranean	Late Iron Age

Archaeological implication: A clear east-west gradient of increasing aridity is detectable in the isotopic data. This aligns with modern climatic patterns and suggests that ancient farmers adapted their agricultural strategies to local environmental constraints.

Finding 4: Chronological trends show increasing water stress over time

Mean $\delta^{13}\text{C}$ values become progressively more positive (increasing water stress) from Neolithic (-24.0‰) to Iron Age (-21.5‰). This could reflect:

- Climate drying during the Bronze and Iron Ages (well documented in Eastern Mediterranean palaeoclimate records)
- Expansion of agriculture into more marginal areas
- Changes in crop selection or irrigation practices

Finding 5: Geography matters as much as chronology

Latitude and longitude together account for 23% of feature importance. The Mediterranean basin classification (Western, Central, Eastern) captures regional agricultural traditions that persist across chronological periods.

Archaeological implication: Regional agricultural practices are deeply embedded in local environmental and cultural contexts. A "one-size-fits-all" model of ancient

agriculture is inadequate.

3.2 Interactive Map

An interactive map was created using **Folium** with a vintage National Geographic base layer (Esri NatGeo_World_Map) to facilitate archaeological interpretation.

Map features:

Feature	Description
Base layer	National Geographic style (sepia/olive tones) for archaeological aesthetic
Markers	Colour-coded by isotopic cluster (brown, sienna, peru, chocolate)
Marker clustering	Density-based grouping with custom colours (<10, 10-50, >50 samples)
Popups	Display site name (Archaeological_Site), cereal type, $\delta^{13}\text{C}$, $\delta^{15}\text{N}$, period, basin
Legend (top-right)	Cluster colours with mean isotopic values
Legend (bottom-centre)	Density ranges for sample aggregation
Scale	Bottom-right corner, repositioned for clarity

Interpretation using the map:

- **Western Mediterranean (Iberia, southern France, Italy west of 11°E):**
Dominated by Clusters 0 and 1 (optimal to moderate conditions). Sites like *Castellaro dell'Uscio* (Italy) show high $\delta^{13}\text{C}$ values consistent with good water availability.
- **Central Mediterranean (Italy east of 11°E, Adriatic, Greece west of 23°E):**
Mixed clusters 1 and 2. Transition zone between western humidity and eastern aridity.
- **Eastern Mediterranean (Aegean, Anatolia, Levant):** Dominated by Clusters 2 and 3 (high stress to extreme aridity). Sites like *Case Bastione* show elevated $\delta^{15}\text{N}$ values suggesting manuring or arid soils.

Example popup from the map:

Empieza a programar o a [crear código](#) con IA.

